DATA SCIENCE
&
ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 2

only for DS/AI

Lecture No. 02



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Recap of previous lecture



Topic

BIVARIATE RANDOM VARIABLE

(Discrete Case = joint p.m.f)

Topics to be Covered



Topic MULTIPLE INTEGRATION

Joint P.M.F / Joint Prob Dist.

① If X & Y are two Random Variables then joint p.m.f is given as

$$p_{ij} = P[x_i, y_j] = P[X = x_i \ \& \ Y = y_j]$$

$$p_{ij} \geq 0$$

$$\sum_{i=1}^{m} \sum_{j=1}^{n} p_{ij} = 1$$

② Individual Prob Dist. of X & Y are called M.D.f defined as,

X :	x_1	x_2	x_3	----	x_m
$P(X)$:	p_1	p_2	p_3	---	p_m

&

Y :	y_1	y_2	y_3	----	y_n
$P(Y)$:	p_1'	p_2'	p_3'	---	p_n'

③ X & Y are called Ind R.V if $\forall$ x_i & y_j we have $P(X = x_i, Y = y_j) = P(X = x_i) \cdot P(Y = y_j)$

Conditional Dist for joint p.m.f is given as;

$$P\left[\frac{X=x_i}{Y=y_j}\right]=\frac{P(x_i,y_j)}{P(Y=y_j)} \quad \& \quad P\left[\frac{Y=y_j}{X=x_i}\right]=\frac{P(x_i,y_j)}{P(X=x_i)}$$

[NAT]

HW8

#Q. Two Dice are rolled simultaneously. Let x denotes the sum of two faces and y the absolute value of their difference. Find joint p.m.f for x and y.

(i) Also evaluate P (x = y) Ans = 0

(ii) P [x = 7 ∩ y ≥ 3] Ans = 1/9

good

(iii) P [(x + y = 4) ∩ (x − y = 2)] = ? Ans = 1/18

(iv) Find marginal distributions of x and y

(v) Find conditional distribution of x given y = 2

$X = \{\text{sum of two Faces}\} = \{2, 3, 4, 5 \ldots, 11, 12\}$

$Y = \{\text{Diff of their outcomes}\} = \{0, 1, 2, 3, 4, 5\}$

$S = \begin{Bmatrix} (11)(12)\ldots(16) \\ (21)(22)\ldots(26) \\ (31)\ldots\ldots(66) \end{Bmatrix}$

$n(S) = 6 \times 6 = 36$ pair

M.D.F for X :⟶

X :	2	3	4	5	6	7	8	9	10	11	12
$P(X)$:	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

M.D.F for Y :⟶

Y :	0	1	2	3	4	5
$P(Y)$:	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$

$Y = 0 \Rightarrow$ Diff $= 0$ Ans

fav $= \{(11)(22)(33)(44)(55)(66)\}$

$Y = 1 \Rightarrow$ Diff $= 1$ Ans

fav $= \begin{Bmatrix} (12)(23)(34)(45)(56) \\ (21)(32)(43)(54)(65) \end{Bmatrix}$

$Y = 2 \Rightarrow$ Diff $= 2$ Ans

fav $= \begin{Bmatrix} (13)(24)(35)(46) \\ (31)(42)(53)(64) \end{Bmatrix}$

Joint Dist Table:-

X \ Y	0	1	2	3	4	5	P(X)
2	1/36	0	0	0	0	0	1/36
3	0	2/36	0	0	0	0	2/36
4	1/36	0	2/36	0	0	0	3/36
5	0	2/36	0	2/36	0	0	4/36
6	1/36	0	2/36	0	2/36	0	5/36
7	0	2/36	0	2/36	0	2/36	6/36
8	1/36	0	2/36	0	2/36	0	5/36
9	0	2/36	0	2/36	0	0	4/36
10	1/36	0	2/36	0	0	0	3/36
11	0	2/36	0	0	0	0	2/36
12	1/36	0	0	0	0	0	1/36
P(Y)	6/36	10/36	8/36	6/36	4/36	2/36	=1

① $P(x=y) = P[(22) \text{ or } (33) \text{ or } (44) \text{ or } 55)] = 0+0+0+0 = 0$

② $P(x=7, y \geq 3) = P[(73) \text{ or } (74) \text{ or } (75)] = \frac{2}{36} + 0 + \frac{2}{36} = \frac{1}{9}$

③ $P[x+y=4 \cap x-y=2] = ? = P[(31)] = \frac{2}{36}$

$x+y=4 \Rightarrow$ fav Cases $= \{(22), (31), (40)\}$ ⟶ $x+y=4 \cap x-y=2 = (31)$

where $x = \{2, 3, 4, \ldots, 11, 12\}$ & $y = \{0, 1, 2, 3, 4, 5\}$

$x-y=2 \Rightarrow$ fav Cases $= \{(20), (31), (42), (53), (64), (75)\}$

(iv) Conditional Dist of X for given $Y=2$ is $P\left[\frac{X=x_i}{Y=2}\right] = \frac{P[(x_i,2)]}{P(Y=2)} = \frac{?}{(8/36)}$

X :	2	3	4	5	6	7	8	9	10	11	12
$P(X=x_i/Y=2)$:	$\frac{(2,2)}{8/36}$	$\frac{(3,2)}{8/36}$	$\frac{(4,2)}{8/36}$	$\frac{(5,2)}{8/36}$	$\frac{(6,2)}{8/36}$	$\frac{(7,2)}{8/36}$	$\frac{(8,2)}{8/36}$	$\frac{(9,2)}{8/36}$	$\frac{(10,2)}{8/36}$	$\frac{(11,2)}{8/36}$	$\frac{(12,2)}{8/36}$

X :	2	3	4	5	6	7	8	9	10	11	12
$P(X=x_i/Y=2)$:	0	0	$\left(\frac{2/36}{8/36}\right)$	0	$\left(\frac{2/36}{8/36}\right)$	0	$\left(\frac{2/36}{8/36}\right)$	0	$\frac{2/36}{8/36}$	0	0

$\Rightarrow$

X :	2	3	4	5	6	7	8	9	10	11	12
$P(X=x/Y=2)$:	0	0	1/4	0	1/4	0	1/4	0	1/4	0	0

INTEGRATION (BASICS)

① $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

② $\int_{-a}^{a} f(x)\,dx = \int_0^a [f(x)+f(-x)]\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(-x)=f(x), \text{ i.e Even func}^n \\ 0, & \text{if } f(-x)=-f(x) \text{ i.e odd func}^n \end{cases}$

③ $\int_0^{2a} f(x)\,dx = \int_0^a [f(x)+f(2a-x)]\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(2a-x)=f(x) \\ 0, & \text{if } f(2a-x)=-f(x) \end{cases}$

Q $I=\int_0^{2\pi}\cos^5 x\,dx=?$ By Prop (3) $=\int_0^{\pi}\left[\cos^5 x+\cos^5(2\pi-x)\right]dx=\int_0^{\pi}\left(\cos^5 x+\cos^5 x\right)dx$

$=2\int_0^{\pi}\cos^5 x\,dx=$ Prop 3 Again use $=2\int_0^{\pi/2}\left[\cos^5 x+\cos^5(\pi-x)\right]dx=2\int_0^{\pi/2}\left(\cos^5 x-\cos^5 x\right)dx$

$=0$

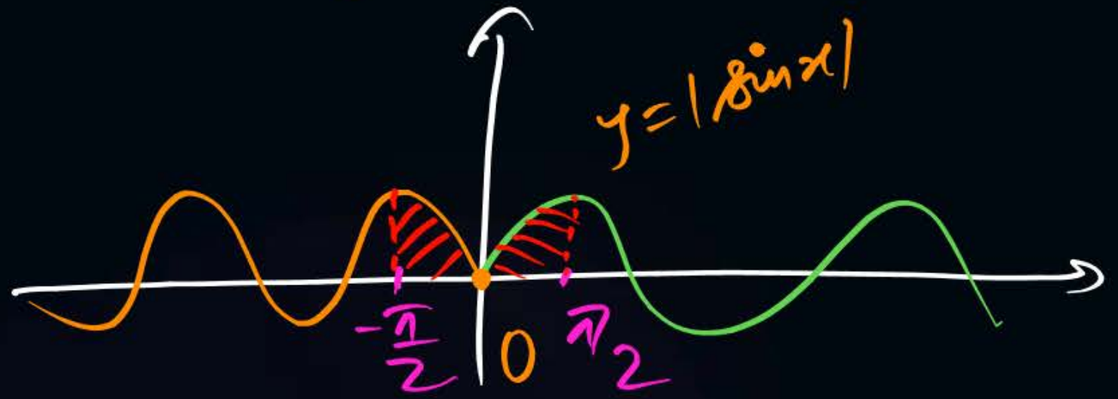
(PQ) $I=\int_0^{2\pi}|\cos x|\,dx=?$ Prop 3 $=2\int_0^{\pi}|\cos x|\,dx$ Prop 3 $=4\int_0^{\pi/2}|\cos x|\,dx=4\int_0^{\pi/2}(+\cos x)\,dx=4(\sin x)_0^{\pi/2}$

$=4(1-0)=4$

Q $I=\int_{-\pi/2}^{\pi/2}(\sin^7 x)\,dx=?$ Prop 2 $=0$ $\left(\because f(x)=\sin^7 x \text{ is an odd func}^n\right)$

Q. $I = \int_{-\pi/2}^{\pi/2} |\sin x|\, dx = ?$ Even $= 2\int_0^{\pi/2} |\sin x|\, dx = 2\int_0^{\pi/2} (+\sin x)\, dx = 2(-\cos x)_0^{\pi/2}$

$= -2(0-1) = 2$



Q. $I = \int_{-1}^{1} \log\left(\frac{2-x}{2+x}\right) dx = ? = 0$

$f(x)$

Now, $f(-x) = \log\left[\frac{2+x}{2-x}\right] = \log\left(\frac{2-x}{2+x}\right)^{-1} = -\log\left(\frac{2-x}{2+x}\right) = -f(x)$ i.e. $f(x)$ is an odd func^n

MULTIPLE INTEGRAL

Double Integral Working Rule →

Case I: If y has variable limits, we should first integrate y keeping x constant.

Case II: if x " " " " " " x " y " "

Case III: if Both the variables have constant limits, we can integrate in any order, But it is advisable to integrate in that order which is given in Question.

If Both the limits are constants OR integrand is an Implicit func^n of x & y then we can integrate in any order keeping other variable constant.

Case IV-e If Both the limits are constants and Integrand is an Explicit func^n of x & y then we can integrate separately.

Q. $I=\int_0^2\int_0^{x^3}(xy^2)\,dy\,dx=?=\int_{x=0}^{2}\int_{y=0}^{x^3}xy^2\,dy\,dx=\int_{x=0}^{2}\left[\int_{y=0}^{x^3}y^2dy\right](x\,dx)$

$=\int_{x=0}^{2}\left[\frac{y^3}{3}\right]_{y=0}^{x^3}.(x\,dx)=\frac{1}{3}\int_{x=0}^{2}x^{10}dx=\frac{1}{3}\left(\frac{x^{11}}{11}\right)_{x=0}^{2}=\frac{1}{3}\left(\frac{2^{11}}{11}\right)$ Ans

Q. $I=\int_0^1\int_{y^2}^{\sqrt{y}}(1)\,dx\,dy=?=\int_{y=0}^{1}\int_{x=y^2}^{\sqrt{y}}(1)\,dx\,dy=\int_{y=0}^{1}(x)_{x=y^2}^{\sqrt{y}}.(dy)=\int_{y=0}^{1}(\sqrt{y}-y^2)\,dy$

$=\left(\frac{y^{3/2}}{3/2}-\frac{y^3}{3}\right)_{y=0}^{1}=\cdots=\frac{1}{3}$

Q $I = \int_0^2 \int_0^{x/2} e^{x^2} \, dy \, dx = ? = \int_{x=0}^{2} \left\{ \int_{y=0}^{x/2} 1 \, dy \right\} (e^{x^2} dx) = \int_{x=0}^{2} \left\{ \frac{x}{2} - 0 \right\} e^{x^2} dx$

$= \frac{1}{4} \int_0^2 (2x) e^{x^2} \underline{dx} = \frac{1}{4} \int_0^4 e^t \, dt = \frac{1}{4} \left[e^t \right]_{t=0}^{4} = \frac{1}{4}(e^4 - e^0) = \frac{e^4 - 1}{4}$

Put $x^2 = t$ → At $x=0$, $t=0$; At $x=2$, $t=4$

$2x \, dx = dt$

Q $\int_0^2 \int_0^x (x^2 y) \, dy \, dx = ? = \int_{x=0}^{2} x^2 dx \times \int_{y=0}^{x} y \, dy = \left(\frac{x^3}{3}\right)_0^2 \times \left(\frac{y^2}{2}\right)_0^x = \frac{8}{3}\left(\frac{x^2}{2}\right) = \frac{4x^2}{3}$

explicit fmcy

WRONG APP ∵ limits are not const.

Correct App- $I = \int_{x=0}^{2} \left(\int_{y=0}^{x} y \, dy \right) x^2 dx = \frac{1}{2} \int_{x=0}^{2} (x^2)(x^2) dx = \frac{1}{2} \left(\frac{x^5}{5} \right)_0^2 = \frac{32}{10} = \frac{16}{5}$

Q. $I=\int_0^3\int_1^2 e^{(x+y)}\,dx\,dy = ? = \int_{y=0}^{3}\int_{x=1}^{2} e^x e^y\,dx\,dy =$ M-I, M-II, M-III

(M-I) $I=\int_{y=0}^{3}\left\{\int_{x=1}^{2} e^x dx\right\}(e^y dy) = (e^2-e)\int_{y=0}^{3} e^y dy = (e^2-e)(e^3-1)$

(M-II) $I=\int_{x=1}^{2}\left\{\int_{y=0}^{3} e^y dy\right\}(e^x dx) = \int_{x=1}^{2}(e^3-1)\,e^x dx = (e^3-1)\int_1^2 e^x dx = (e^3-1)(e^2-e)$

(M-III) $I=\int_{y=0}^{3}\int_{x=1}^{2} e^x\cdot e^y\,dx\,dy = \int_{x=1}^{2} e^x dx \times \int_{y=0}^{3} e^y dy = (e^2-e)(e^3-1)$ Ans

Explicit func^n.

Q: $I=\int_0^{\pi/2}\int_0^{\pi/2}\sin(x+y)\,dx\,dy=?=\int_{y=0}^{\pi/2}\int_{x=0}^{\pi/2}\sin(x+y)\,dx\,dy$ (Implicit form)

M-I / M-II

(M-I) $\int_{y=0}^{\pi/2}\left\{\int_{x=0}^{\pi/2}\sin(x+y)\,dx\right\}(dy)=\int_{y=0}^{\pi/2}\left\{-\cos(x+y)\right\}_{x=0}^{\pi/2}(dy)=-\int_{y=0}^{\pi/2}\left\{\cos\left(\frac{\pi}{2}+y\right)-\cos y\right\}dy$

$=\int_{y=0}^{\pi/2}(\sin y+\cos y)\,dy=(-\cos y+\sin y)_{y=0}^{\pi/2}=\ldots\ldots=2$

(M-II) $I=\int_{x=0}^{\pi/2}\left\{\int_{y=0}^{\pi/2}\sin(x+y)\,dy\right\}(dx)=\ldots\ldots\ldots\ldots=2$

Triple Integration Working Rule →

Case I: if all the variables have constant limits then we can integrate in any order keeping other variables constant

Case II: if one or two variables have variable limits then

"" which one is more variable should be integrated first ""

i.e. $$I=\int_a^b\int_{f(x)}^{g(x)}\int_{\phi(x,y)}^{\psi(x,y)} f(x,y,z)\,dx\,dy\,dz = ? = \int_a^b\int_{f(x)}^{g(x)}\int_{\phi(x,y)}^{\psi(x,y)} f(x,y,z)\,dz\,dy\,dx$$

$x=$ $y=$ $z=$

Applications of Multiple Integral

① $\boxed{\text{Area} = \iint (1)\, dx\,dy} = \int y\,dx = \int_{x=a}^{b} f(x)\,dx$



② $\boxed{\text{Volume} = \iiint (1)\, dx\,dy\,dz} = \iint z\,dx\,dy = \iint f(x,y)\,dx\,dy$

③ $\text{Mass} = \iiint (\rho)\, dx\,dy\,dz$

Ex $I = \int_0^{2\pi}\int_0^{\pi/3}\int_0^{1} x^2 \sin y \, dx\,dy\,dz = ? = \int_{z=0}^{2\pi}\int_{y=0}^{\pi/3}\int_{x=0}^{1} (x^2 \sin y)\,dx\,dy\,dz$

Explicit func^n

$= \int_{x=0}^{1} x^2\,dx \times \int_{y=0}^{\pi/3} \sin y\,dy \times \int_{z=0}^{2\pi} (1)\,dz$

$= \left(\frac{x^3}{3}\right)_{x=0}^{1} \left(-\cos y\right)_{y=0}^{\pi/3} \left(z\right)_{z=0}^{2\pi}$

$= \left(\frac{1}{3}-0\right)\left(-\left\{\frac{1}{2}-1\right\}\right)\left(2\pi-0\right) = \frac{\pi}{3}$

Q. $I = \int_0^1 \int_0^{z} \int_0^{x+z} (1)\, dx\, dy\, dz = ? = \int_{z=0}^{1} \int_{x=0}^{z} \int_{y=0}^{x+z} (1)\, dy\, dx\, dz$

$= \int_{z=0}^{1} \int_{x=0}^{z} (x+z)\, dx\, dz = \int_{z=0}^{1} \left\{ \frac{x^2}{2} + zx \right\}_{x=0}^{z} (dz) = \int_{z=0}^{1} \left(\frac{z^2}{2} + z^2 \right) dz$

$= \frac{3}{2} \int_{z=0}^{1} z^2\, dz = \frac{3}{2} \left(\frac{z^3}{3} \right)_{z=0}^{1} = \frac{1}{2} = 0.5$

Doubt by student

P.T.O

Telephone calls arrive at an exchange according to the Poisson process at a rate $\lambda =$ 2/min. Calculate the probability that exactly two calls will be received during each of the first 5 minutes of the hour.

Method II

$N = \{$No. of calls per min$\}$

$\lambda = 2$ calls/min

$P(N=2 \text{ calls}) = \frac{e^{-\lambda}\lambda^{r}}{r!} = \frac{e^{-2}2^{2}}{2} = \frac{2}{e^{2}}$

Method I

$X = \{$Number of minutes in which two calls are coming$\}$

$p = \frac{2}{e^2}$, $q = 1 - \frac{2}{e^2}$, $n = 5$

$P(X = r \text{ success}) = {}^{n}C_{r} p^{r} q^{n-r}$

$P(X = 5 \text{ success}) = {}^{5}C_{5} (p)^{5} (q)^{0} = \left(\frac{2}{e^2}\right)^{5}$

So Req Prob $= \left(\frac{2}{e^2}\right)^{5}$ Ans

(M-II) using poisson $p = \frac{2}{e^2}$

$$\text{Req Prob} = \left(\frac{2}{e^2}\right)\left(\frac{2}{e^2}\right)\left(\frac{2}{e^2}\right)\left(\frac{2}{e^2}\right)\left(\frac{2}{e^2}\right)$$

1st 2nd 3rd 4th 5th

$$= \left(\frac{2}{e^2}\right)^5$$

THANK - YOU